23 · Myopia, hyperopia and astigmatism

The words of this track, as it says them. Bold brown is the examiner.

Myopia, hyperopia and astigmatism. Five marks.

Explain how myopia, emmetropia and hyperopia differ, and how each error is corrected. What is astigmatism?

The hook: myopia, my world is up close. Short-sighted. The focus falls short, in front of the retina, and a concave lens pushes it back. Hyperopia: the focus overshoots, behind the retina, and a convex lens pulls it forward.

Emmetropia is normal vision. Parallel light rays are focused on the retina.

Myopia is short-sightedness. Light from distant objects focuses in front of the retina, usually because the eyeball is too long. Near things are clear, and far things are blurred. It's corrected with a concave lens, which diverges the light, spreading it out, and moves the focus back onto the retina.

Hyperopia is long-sightedness. Light from near objects focuses behind the retina, usually because the eyeball is too short. Far things are clearer than near ones. It's corrected with a convex lens, which converges the light and moves the focus forward onto the retina.

Astigmatism: the cornea or the lens is curved unevenly, more in one direction than another. So light focuses at several points, and vision is blurred at all distances. It's corrected with glasses or contact lenses, or with laser surgery.

The traps. Myopia: short-sighted, concave. Hyperopia: long-sighted, convex. Swapping the lenses is the commonest slip. Second: say where the image falls, in front of or behind the retina. That's the mark, not just "blurry". Third: astigmatism doesn't depend on distance. It's blurred near and far.

Once more. Myopia, hyperopia and astigmatism.

Emmetropia: light is focused on the retina. Myopia, short-sighted: distant light focuses in front of the retina because the eyeball is too long, and a concave lens diverges the light. Hyperopia, long-sighted: near light focuses behind the retina because the eyeball is too short, and a convex lens converges the light. Astigmatism: the cornea or lens is curved unevenly, so light focuses at several points and everything is blurred. Glasses, contacts, or laser surgery.